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1 Project description

The project for this course is to develop a risk calculation system.

The homework assignments have included (and will continue to include) various components of this system. You will need to assemble these pieces into a coherent whole. You will need to choose models and methodologies, document and justify the choices and write and test the code.

2 Requirements

The risk calculation system should be able to:

- Take a portfolio of stock and option positions as input.
- Both calibrate to historical data and take parameters as input.
 - Historical data can be supplied as part of the input data.
- Compute Monte Carlo, historical, and parametric VaR.
- Compute Monte Carlo and historical ES.
- Backtest the computed VaRs against history.

The requirements are intentionally vague. In practice you will be faced with extremely vague requirements specifications and you will need to exercise judgment as to the best way to reasonably fulfill the given requirements. The expectation is that the course is giving you the background necessary to make such judgment calls.

It should be relatively easy with your system to take an arbitrary set of stocks and options as input and do the above calculations. One way of doing this is by creating a standardized input file format and writing software to read the input file and perform the requested calculations. This would be the only way to proceed if working in C++. If working in an interpreted language (like matlab or R) one could alternatively provide the appropriate code so that these calculations can be done by writing small scripts.

3 Deliverables

The project should follow sound model risk management practices. As such, it has the following 5 deliverables:

1. Model documentation.
2. Software design documentation.
3. Test plan.
4. Software.
5. Test results.

The details of what constitutes appropriate model documentation, software design documentation and test plans were covered in lecture 5.

For a sample of model documentation, see *Model Validation Municipal Bonds*, Stein [Ste14a]. A template for model documentation can be found in *Model Validation Report Template*, Stein [Ste14b]. Both documents are available in the references directory.

4 Grading

The 5 deliverables will be graded as in Table 1. This gives 40% to the software (parts 2 and 4), 30 points to the testing (3 and 5), and 30 points to the model documentation (part 1).

Deliverable	Points
1. Model documentation	30
2. Software design documentation	15
3. Test plan	20
4. Software	25
5. Test results	10

Table 1: Project grading rubric.

Deliverable 1 will be graded on base on:

- How complete it is
- How clear it is
- How well it documents and justifies all of the modeling choices and their limitations (as per the model validation template class and lecture 5).

Points will be taken off for poor modeling decisions (like not modeling changes in volatility for options, or using historical volatility instead of inputting implied volatility to price options). Points will also be taken off if the testing (deliverables 3 and 5) do not support the conclusions here. Points will also be taken off

for modeling errors, such as not calculating covariance correctly, inappropriate choices for parametric VaR, etc.

Deliverable 2 will be graded based on how clear and complete the design documentation is. Discussion of what this should include is also in the model validation lecture.

Deliverable 3 will be graded based on how complete the test suite is that validates the model performance. Deliverable 5 gets graded based on how accurately it carries out the test plan and records and evaluates the results.

Deliverable 4 will be graded based on how complete, correct, and well written the software is. Less functionality, poorer code, and poorer code documentation will get a lower grade than more comprehensive, well documented software. To get full points, the software has to be able to handle an arbitrary set of securities, allow the user to provide the appropriate histories (for all calculations), or the parameters to use (for Monte Carlo and parametric calculations), do ES and VaR calculations with all models, and do back testing of VaR.

References

- [Ste14a] Harvey J. Stein. *Model Validation Municipal Bonds*. 2014.
- [Ste14b] Harvey J. Stein. *Model Validation Report Template*. 2014.